



An adaptive, secure and imperceptive image watermarking using swarm intelligence, Arnold transform, SVD and DWT

Falgun Thakkar¹ • Vinay Kumar Srivastava²

Received: 24 September 2019 / Revised: 18 October 2020 / Accepted: 9 December 2020 /

Published online: 9 January 2021

© Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract

Digital image watermarking is one of the important method of copyright protection and rightful ownership of the digital image and video. In this paper, the existing algorithm based on DWT-SVD is improved and made adaptive with particle swarm optimization (PSO) which increases the robustness and imperceptibility. Existing as well as proposed algorithm based on PSO creates diagonal line in the extracted watermark image against some of the noise attacks. Further to remove diagonal line from the extracted watermark image an algorithm based on PSO and Arnold transform is proposed. Both proposed algorithms (PSO and AT-PSO) are implemented and tested with different types of cover images. The comparative analysis of these two proposed algorithms is carried out with the existing algorithm. The comparative analysis shows that under most of noise attacks both the proposed algorithms outperform over the existing algorithm in terms of Peak Signal to Noise Ratio (PSNR) and Normalized Similarity Ratio (NSR). Moreover, the imperceptibility offered by Arnold transform based algorithm is better as compared to the other proposed algorithm. Robustness of both the proposed algorithms is close to each other.

Keywords Adaptive image watermarking · Particle swarm optimization · Scrambling · Security · Normalized similarity ratio

✉ Falgun Thakkar
falgunget@gmail.com

Vinay Kumar Srivastava
vinay@mnnit.ac.in

¹ Electronics and Communication Engineering Department, G H Patel College of Engineering and Technology, V.V.Nagar, Gujarat 388120, India

² Electronics and Communication Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, Uttar Pradesh, India

1 Introduction

The watermarking scheme should provide enough imperceptibility, trustworthiness and robustness to protect the secret watermark from various noise attacks and offer proof for protection of rightful ownership [18, 22, 29]. These watermarking schemes are divided into two different domains: spatial-domain and frequency-domain. The spatial domain watermarking is less complex and easily implementable and mostly used as fragile watermarking techniques. Contrary, the frequency domain watermarking are the computationally complex and used as a robust watermarking techniques [17].

To protect the copyright information in network environment the watermarking techniques are proposed [5, 22]. The Singular Value Decomposition (SVD) based watermarking algorithms are implemented to get better robustness under variety of noise attacks [1–4, 14, 15, 17, 18, 21–23, 28, 29, 32]. The SVD based techniques proposed by [22] is implemented by [30, 36] and commented that the given technique is not protecting the rightful ownership. The reliable watermarking technique based on Principle Components (PCs) is proposed in [14] which remove the problem of rightful ownership associated with SVD based technique but this PCs based technique is more sensitive to any noise attack. This limitation of PCs based technique is removed using optimization techniques like Particle Swarm Optimization (PSO) and Differential Evaluation (DE) in [1, 28, 29]. The SVD, Discrete Wavelet Transform (DWT) and Chebyshev chaotic map based watermarking technique is given in [32] for the rightful ownership protection. The Integer Wavelet Transform (IWT) based watermarking technique is used to get better robustness is presented in [23]. The watermark is embedded three times into the SVD blocks of cover image to get better security and robustness [21]. The watermark image is directly embedded inside the singular values of DWT coefficients of cover image to get better imperceptibility and robustness [15]. To increase security, robustness and rightful ownership the non-overlapping block based SVD with basic watermarking algorithm [7, 8, 13, 25] is used. Different transforms like Binomial, Stirling and Lah are used in [9–11] for watermarking and security of image data to archive higher payload.

Many evolutionary and swarm intelligent based algorithms like Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) are used to solve different types of optimization problems. Essentially they are probabilistic type and solve optimization problems using variety of controlling parameters like population size, elite size and number of iteration or generation etc. These parameters are varied from one type of algorithm to other [27]. For example, GA uses probability of mutation and crossover along with selection parameters. PSO uses inertia weight as well as social and cognitive parameters. To optimize robustness and fidelity of watermarked image the Genetic Algorithm (GA) is used in [12, 31] whereas the PSO based watermarking technique is proposed in [34] to increase the robustness and imperceptibility. The Arnold Transform (AT) based watermarking technique in the Discrete Cosine Transform (DCT) and DWT domains are given in [19, 26, 33].

It is found that the algorithm given in [15] suffers from lower robustness and imperceptibility due to improper method is used to select the scale factor. The diagonal line is also present in the extracted watermark image. In the scheme proposed in this paper, the scale factor selection methodology is improved with two different approaches. The first approach is based on the PSO which improves the robustness and imperceptibility but the diagonal line is still present in the extracted watermark. To remove this diagonal line from extracted watermark and increase security, the Arnold Transform (AT) along with PSO based technique is proposed. This technique also increases the imperceptibility of the algorithm.

The rest of the paper is organized as follows. Section 2 describes main contribution of this paper, preliminaries on PSO, AT, SVD and DWT is given in section 3. The proposed watermark embedding and extraction algorithms are presented in section 4. The performance measures like PSNR, Normalized Correlation Coefficient (NCC) and Normalized Similarity Ratio (NSR) is given in section 5. In section 6, the experimental results are presented and discussed. In the last, conclusions about the proposed techniques are given.

2 Main contribution of the work

In the proposed work, SVD is applied on the low frequency wavelet coefficients of cover image. A scaled pixel values of watermark image are embedded and modified the singular values of selected wavelet coefficients. A scale factor is decided by the evolutionary algorithm PSO. Further, SVD is applied on these modified singular value matrix. A modified low frequency wavelet coefficients set is generated using inverse singular value decomposition (ISVD) applied on generated singular values along with orthogonal matrices of low frequency wavelet coefficients of cover image.

Adaptive to different groups of images In existing schemes like [15] and others, the scale factor selection is manual and based on cover and watermark image dependent. To eliminate this limitation of such schemes, PSO is incorporated which is suggested a suitable scale factor for given pair of cover and watermark image in this proposed work.

Free from False positive problem In most SVD based watermarking algorithms [5, 22], a false positive problem is common which is reported in [35]. Here it is shown that, extraction of original watermark was possible using singular values of image than original one. This problem has been eliminated by proposed algorithm of this paper using dual SVD of selective components of cover image.

Removal of diagonal Line and Security Assurance As singular value modification based watermarking algorithms are embedded the singular values of watermark content into the Host image, the diagonal line is visible in extracted watermark content. In this proposed scheme use of Arnold Transform leads to eliminate a diagonal line from extracted watermark. Moreover, AT is also offered secure information transmission using this proposed algorithm [6, 20].

3 Preliminaries

3.1 Particle swarm optimization

A PSO has been used due to its certain attractive features like ease of implementation and it can optimum solution of problem without gradient information. Nowadays, it is rigorously used to solve neural network training and function minimization types of wide array problem [34]. In addition to that, tuning of the PSO based algorithm require very few parameters as compared to other Heuristic algorithms which leads to its faster convergence with higher accuracy [16]. It is also used in image processing applications like compression and watermarking.

The PSO is initialized with random particles in respective search space. These random particles are tuned on the search space by adjusting their velocities [16]. Each particle have two information, first is its own best experience (position, P_i and objective value, P_{best}) and best experience from whole swarm (position, P_g and objective value, P_{best}). The mathematical formulation of velocity and position is given below

$$x_i = (x_{i1}, x_{i2}, \dots, x_{in}) \quad (1)$$

$$v_i = (v_{i1}, v_{i2}, \dots, v_{in}) \quad (2)$$

These positions, x_i and velocity, v_i are updated using following Eqs. (3) and (4) and fitness function value is checked with its best value. This process is repeated until the desired value of fitness function is achieved.

$$v_{im}(n+1) = v_{im}(n) + c_1 r_{1m}(p_{im} - x_{im}) + c_2 r_{2m}(p_{gm} - x_{im}) \quad (3)$$

$$x_{im}(n+1) = x_{im}(n) + v_{im}(n+1) \quad (4)$$

Where C_1 and C_2 are acceleration coefficients numbers whereas the r_{1m} and r_{2m} are uniformly distributed random number in the range $[0, 1]$. A flow chart of PSO which executed in proposed work of this paper is depicted in following Fig. 1.

3.2 Singular value decomposition

Singular value decomposition (SVD) is a tool which is used in various numerical applications. It can partition any rectangular or square matrix M into three matrices; U and V are the orthogonal matrices whereas S is the diagonal matrix which contains singular values of original matrix M [1, 23]. The mathematical formulation for the same is given in following equation:

$$M = USV^T \quad (5)$$

The SVD is widely used in various image processing applications due to its some of the characteristics;

- If small perturbation is added to singular value, S there is no significant change in visual quality of image.
- The singular value S gives luminance detail of image whereas the diagonal matrices U and V contain the geometrical detail.

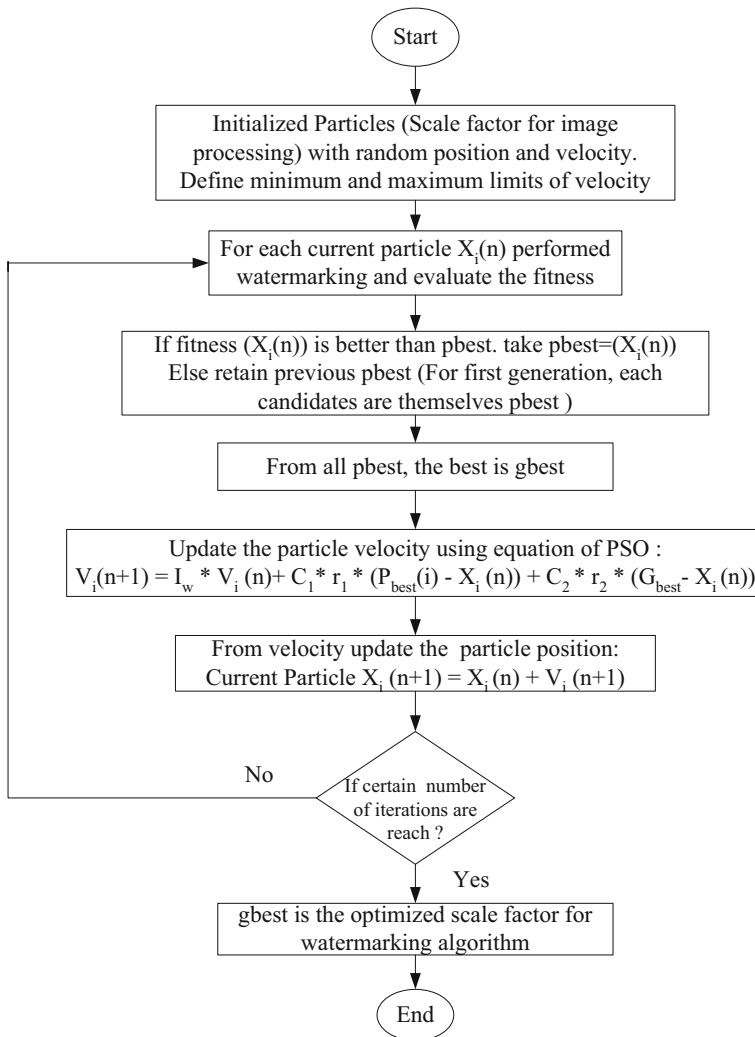


Fig. 1 Particle swarm optimization

$$A = \begin{pmatrix} u_{1,1} & \cdots & u_{1,M} \\ u_{2,1} & \cdots & u_{2,M} \\ \vdots & \ddots & \vdots \\ u_{M,1} & \cdots & u_{M,M} \end{pmatrix} \begin{pmatrix} s_{1,1} & 0 & 0 \\ 0 & s_{2,2} & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & s_{M,N} \end{pmatrix} \begin{pmatrix} v_{1,1} & \cdots & v_{1,N} \\ v_{2,1} & \cdots & v_{2,N} \\ \vdots & \ddots & \vdots \\ v_{N,1} & \cdots & v_{N,N} \end{pmatrix}^T$$

3.3 Arnold transform

The Arnold Transform (AT) is the scrambling technique which rearranges the pixels of image such a way that the original image looks like disorganized and encrypted image. The AT on two dimensional digital image with dimension $N \times N$ is define as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \pmod{N} \quad (6)$$

Here, the x and y are the pixel positions of the digital image whereas the x' and y' are the pixel positions of scramble image as a result of AT [23, 33, 35].

3.4 Discrete wavelet transform

The Discrete Wavelet Transform (DWT) offers good energy compaction property and multi resolution analysis; therefore it is widely used in various image processing applications. The DWT decompose the signal into different frequency components. At each level DWT decomposes the image into four different frequency subbands. These subbands named as lower frequency approximate subband (LL) and three other high frequency subbands: horizontal (HL), vertical (LH) and diagonal (HH). At higher level, wavelet transform further decomposes each LL subband into four different subbands as mention above. Low pass filtering in both Horizontal and vertical direction on image generates the LL subband. The low pass and high pass filtering in two different directions produced LH or HL subbands. The high pass filtering in both directions gives the HH subband. This is depicted in following Fig. 2.

4 Proposed work

In different algorithms like [15] and others, there is a need of optimization to overcome the manual efforts in selection of proper values. To make it possible, PSO based technique is incorporated in this manuscript. Here as per the flow chart of the algorithm shown in Fig. 3, two techniques are proposed namely PSO and AT-PSO are proposed. In first method, tuning of the PSO technique is carried out in such a way that it can provide optimum performance in terms of not only PSNR but also NSS and NCC. The security feature is also incorporated in second method using AT to maintain rightful ownership.

It is depicted in Fig. 3 that, proposed algorithm is embedding the watermark content into the singular values of LL subband of the cover image. A weight factor (α) is chosen using PSO optimization algorithm due to its advantages like fast convergence and capability to tune an algorithm with a few parameters. An inverse

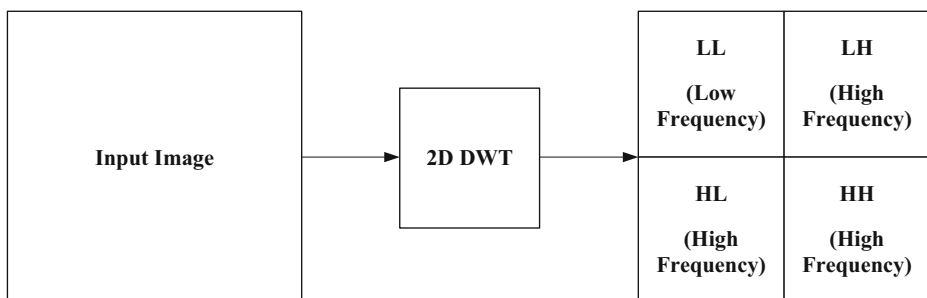


Fig. 2 Single level discrete wavelet transform and respective subbands

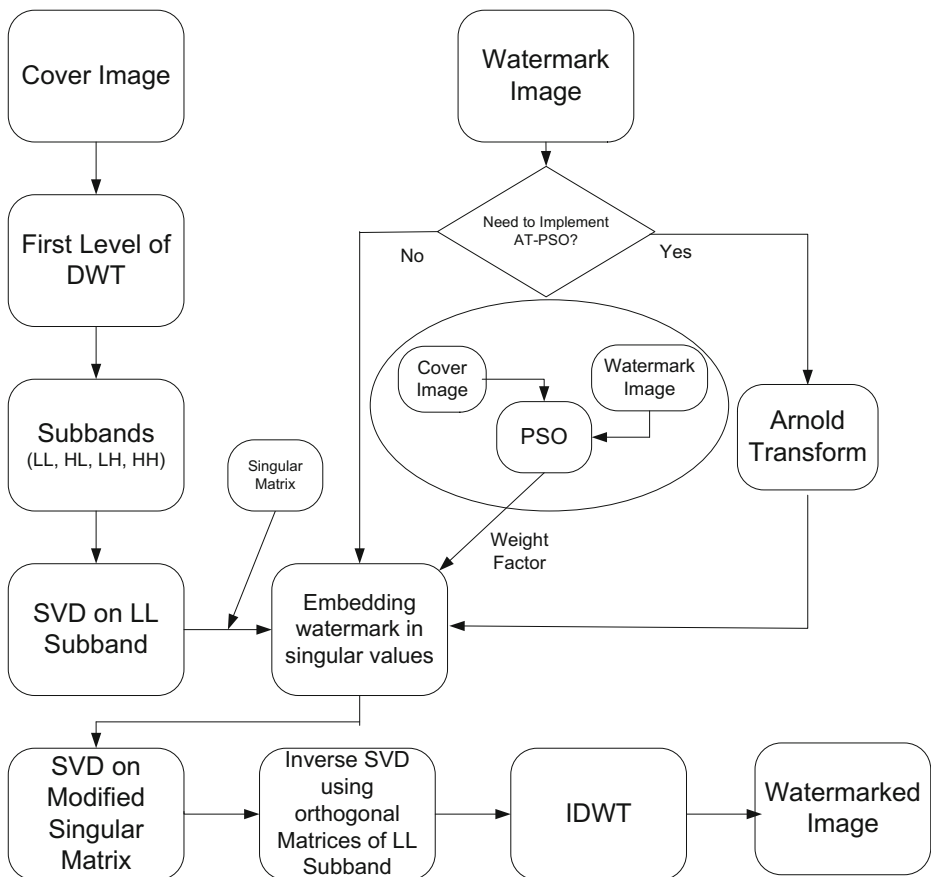


Fig. 3 Flow chart for proposed algorithms PSO and AT-PSO

transform is carried out in terms of SVD and DWT after embed the watermark image successfully into the singular matrix of LL subband'. This first proposed algorithm is named as Image watermarking using PSO optimization technique. In Second proposed algorithm, an AT is incorporated with First proposed algorithm and thus it is termed as AT-PSO based image watermarking. Second algorithm offers high security without much computational complexity over first algorithm. The selection of method is given with decision making symbol in the flow chart.

4.1 Watermark embedding algorithm

In technique proposed this paper, the Arnold Transform scramble the watermark image which is embedded inside the singular value matrix of LL subband of cover image. This embedding process uses PSO to select weight (scale) factor. SVD is applied on this LL subband. Singular value matrix is modified with scramble watermark image and then using inverse SVD the modified LL subband is generated. Inverse DWT is applied on this modified LL subband along with remaining subband

generates the watermarked image. The block diagram for the same is given in Fig. 4 and mathematical steps are given below.

Step 1. Decompose the cover image C into LL , HL , LH and HH subbands using DWT.

Step 2. Apply SVD on the LL subband;

$$U_{LL}S_{LL}V_{LL} = LL \quad (7)$$

Step 3. AT is applied on the watermark image to generate scrambled image W .

Step 4. Modify the singular value matrix S_{LL} with W image using a scale factor α selected by PSO;

$$S_{LLM} = S_{LL} + \alpha * W \quad (8)$$

Step 5. Now, apply the SVD on the Modify the singular value matrix S_{LLM} ;

$$U_{LLMA}S_{LLMA}V_{LLMA} = S_{LLM} \quad (9)$$

Step 6. The Inverse SVD applied on S_{LLMA} with orthogonal matrices U_{LL} and V_{LL} of LL subband, gives modified LL' subband;

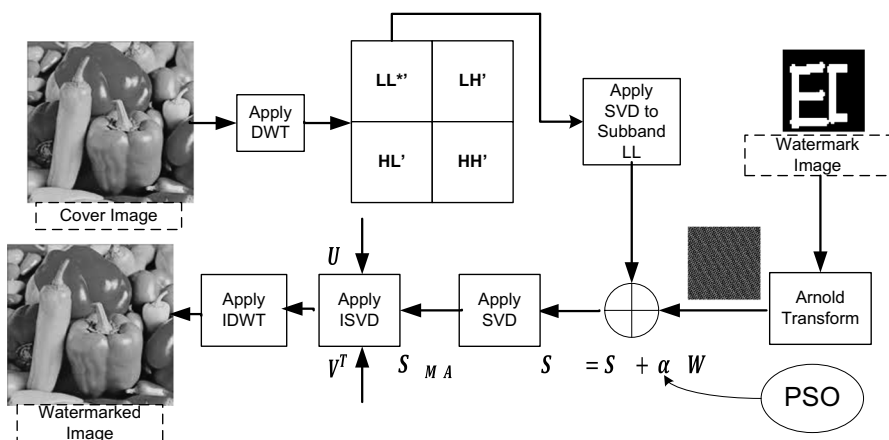


Fig. 4 Proposed scheme for watermark embedded into cover image

$$LL' = U_{LL} S_{LLMA} V_{LL}^T \quad (10)$$

Step 7. The Inverse DWT of the modified LL' subband along with other high frequency subband LH , HL and HH gives the watermarked image C_w .

4.2 Watermark extraction algorithm

The watermarked image C_w may suffer from various geometric and non-geometric type of noise attacks. The mathematical steps to extract the watermark from this noisy watermarked image C_w^* is given below whereas the block diagram is given in Fig. 5.

Step 1. Decompose the noisy watermarked image into LL^* , HL^* , LH^* and HH^* subbands using DWT. Here subband LL^* is the modified and noisy subband.

Step 2. Apply SVD on LL^* subband;

$$U'_{LL} S'_{LLMA} V'_{LL} = LL'^* \quad (11)$$

Step 3. The inverse SVD is applied on respective orthogonal matrices along with S_{LLMA}' to get S_{LLM} ;

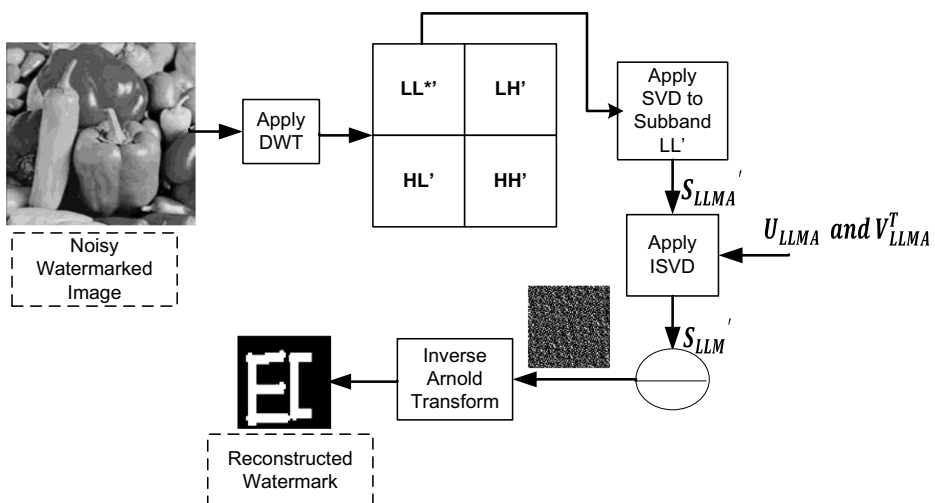


Fig. 5 Proposed scheme for watermark extraction from noisy watermarked image

$$S'_{LLM} = U'_{LLMA} S'_{LLMA} V'^T_{LLMA} \quad (12)$$

Step 4. The scrambled watermark image W' is reconstructed with $S_{LLM'}$ and S_{LL} :

$$W' = (S'_{LLM} - S_{LL}) / \alpha \quad (13)$$

Step 5. The inverse AT applied on the scrambled watermark W' to reconstruct the original watermark image with threshold value 0.5. This threshold value decides the value of pixel 0 or 1.

The mathematical formulation of PSO to get the proper value of scale factor is given below.

The acceleration coefficients C1 and C2 are 2,
 The lower and upper velocities are 10 and 150,
 The maximum iterations are 8.
 The total number of particles is five.

To maximize the imperceptibility and robustness of watermarked image and extracted watermark image the following fitness function [28] is used.

$$f = \max \left[\frac{1}{R} \left(\sum_{i=1}^R \text{corr}_i(w, w') \right) + \text{corr}(c, c_w) \right] \quad (14)$$

In this above equation, the total number of attacks is R whereas the Corr is the Normalized Correlation Coefficient.

As discussed above, It is required to select appropriate parameters for PSO to get better performance. The above parameters like C1, C2, Vmax, Vmin, Iterations and total number of particles are chosen such a way that PSO can maximally optimized the solution. Thus, in a way it is parameter selection impose certain limitations in of the algorithm in which i. e. inappropriate parameters values may leads erroneous output.

5 Performance measures

The robustness of proposed algorithm is verified using Normalized Correlation Coefficient (NCC) and Normalized Similarity Ratio (NSR) whereas imperceptibility is judged using Peak Signal-to-Noise Ratio (PSNR).

The NSR is defined by Eq. (15) which depends on Similarity ratio (SR).

$$NSR = \frac{SR - \min(SR)}{1 - \min(SR)} \quad (15)$$

$$SR = \frac{S}{S + D} \quad (16)$$

The SR is defined in Eq. (16) in which S is total number of similar pixels whereas the D is total number of dissimilar pixels in the original and extracted watermark [15]. The min(SR) is the value of SR which can be achieved for original binary watermark image with respect to an image containing either all white or all black pixels. The NCC and PSNR are defined as following Eqs. (17) and (18) respectively.

$$NCC = \frac{\sum_{i=1}^N \sum_{j=1}^N (w_{ij} - \bar{w}) (w_{ij}^* - \bar{w}^*)}{\sqrt{\sum_{i=1}^N \sum_{j=1}^N (w_{ij} - \bar{w})^2} \sqrt{\sum_{i=1}^N \sum_{j=1}^N (w_{ij}^* - \bar{w}^*)^2}} \quad (17)$$

$$PSNR = 10 \log_{10} \frac{255^2}{MSE} \quad (18)$$

Where, $MSE = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N [C(i, j) - C^*(i, j)]^2$

In Eqs. (17) and (18), the w is original watermark image and w* is the extracted watermark image whereas the C indicates cover image and C* is the watermarked image.

6 Results and discussion

The performance of proposed technique is verified using four different cover images and EC logo as the watermark image, they are depicted in Fig. 6.

The performance analysis in terms of NCC, NSR and PSNR of both the proposed algorithms i.e. PSO based and PSO-AT based algorithms are given in the Tables 1 and 2 respectively. In existing algorithm [15] the selection procedure of scale factor to embed the watermark is empirical and limited up to specific images. Therefore, the NSR of extracted watermark and PSNR of noisy watermarked images are not satisfactory. To improve the existing technique and make it applicable to any type of cover image, the PSO is incorporated in the proposed scheme.

The diagonal line problem is present in the extracted watermark image with existing technique as well as the PSO based proposed technique as shown in the Fig. 7. The Arnold Transform (AT) is included in the PSO based proposed technique which not only remove the diagonal line from extracted watermark image but also increases the imperceptibility of the noisy watermarked image. The NCC and NSR value of the extracted watermark image using both the proposed techniques is above 0.99 against all the types of noise attacks as given in



Fig. 6 The cover images **a** Pepper **b** Elaine **c** Lena **d** Baboon **e** Goldhill **f** Barbara and **g** Cameraman with dimension 512×512 , and the watermark image **h** EC logo with dimension 256×256

Tables 1 and 2. The PSNR values of noisy watermarked images are also better under all the noise attacks except rotation and cropping.

The imperceptibility and robustness of PSO based proposed algorithm and AT-PSO based proposed algorithm are compared with existing algorithm [15] in Tables 3 and 4 for Pepper and Goldhill cover images respectively. In Table 3, the NSR and PSNR values offered by both proposed algorithms are better as compare to the existing algorithm under all the types of noise attacks except rotation and histogram equalization attacks where the PSNR values of noisy watermarked image are slightly less than the existing algorithm. The graphical analysis in Figs. 8 and 9 show the comparison of NSR and PSNR values between existing and AT-PSO based proposed algorithms for Pepper cover image. In Tables 3 and 4, it is also noticeable that the imperceptibility of AT-PSO based proposed algorithm is better than only PSO based proposed algorithm but it is less robust in terms of NSR.

Table 1 The NCC and NSR values of extracted watermark image and PSNR values of noisy watermarked image against various noise attacks for different cover images watermarked with proposed algorithm using PSO

Cover Image/ Attack		Gaussian Blur	Scaling (50%)	Histogram Equi.	Rotation (20°)	JPEG (Q= 25)	Cropping (12%)	Salt & Pepper
Pepper	NCC	0.9999	0.9990	0.9980	0.9973	0.9998	0.9984	0.9962
	NSR	0.9999	0.9985	0.9972	0.9961	0.9997	0.9978	0.9946
	PSNR	39.71	47.60	19.97	9.96	33.38	15.88	22.19
Goldhill	NCC	1	0.9988	0.9938	0.9923	1	0.9979	0.9928
	NSR	1	0.9983	0.9912	0.9890	0.9999	0.9971	0.9898
	PSNR	32.8381	40.1569	17.3367	11.4958	32.9565	19.0697	25.2565
Lena	NCC	0.9980	0.9979	0.9931	0.9979	0.9997	1	0.9954
	NSR	0.9972	0.9970	0.9902	0.9971	0.9995	1	0.9935
	PSNR	40.7982	34.1448	19.1110	10.9721	33.5097	13.5875	23.9002
Elaine	NCC	0.9990	0.9990	0.9938	0.9933	0.9997	1	0.9952
	NSR	0.9986	0.9985	0.9912	0.9905	0.9996	1	0.9932
	PSNR	44.5712	45.7933	18.0856	11.1756	36.1697	12.8869	23.6799

Table 2 The NCC and NSR values of extracted watermark image and PSNR values of noisy watermarked image against various noise attacks for different cover images watermarked with proposed algorithm using PSO and AT

Image/Attack		Gaussian Blur	Scaling (50%)	Histogram Equi.	Rotation (20°)	JPEG (Q= 25)	Cropping (12%)	Salt & Pepper
Pepper	NCC	0.9982	0.9983	0.9964	0.9977	0.9988	0.9972	0.9967
	NSR	0.9974	0.9976	0.9949	0.9968	0.9984	0.9961	0.9953
	PSNR	45.05	41.19	46.27	10.15	38.31	14.59	25.09
Goldhill	NCC	0.9998	0.9978	0.9949	0.9965	0.9985	0.9993	0.9968
	NSR	0.9997	0.9968	0.9928	0.9950	0.9979	0.999	0.9954
	PSNR	36.2169	41.1498	17.3214	10.7319	36.6571	13.7281	25.1117
Lena	NCC	0.9977	0.9978	0.9901	0.9972	0.9987	0.9989	0.9974
	NSR	0.9967	0.9968	0.9931	0.9960	0.9982	0.9985	0.9963
	PSNR	41.1063	33.3386	19.1355	11.0962	33.3878	13.3613	24.9763
Cameraman	NCC	0.9991	0.9999	0.9975	0.9944	0.9995	0.9984	0.9964
	NSR	0.9967	0.9996	0.9910	0.9891	0.9974	0.9943	0.9870
	PSNR	16.6998	34.9976	31.5056	35.2828	31.7383	30.2140	21.0598
Barbara	NCC	0.9978	0.9838	0.9931	0.9847	0.9847	0.9848	0.9858
	NSR	0.9919	0.9714	0.9759	0.9450	0.9452	0.9455	0.9490
	PSNR	24.5254	14.36	29.1040	14.5581	15.3333	14.7683	14.9440
Baboon	NCC	0.9981	0.9979	0.9963	0.9593	0.9982	0.9978	0.9988
	NSR	0.9933	0.9923	0.9868	0.8535	0.9935	0.9921	0.9957
	PSNR	18.1735	20.3159	36.8244	28.9339	17.7073	27.1106	18.7974
Elaine	NCC	0.9982	0.9981	0.9953	0.9975	0.9988	0.9993	0.9974
	NSR	0.9974	0.9973	0.9934	0.9964	0.9984	0.9990	0.9963
	PSNR	41.8383	45.0674	18.0689	11.1666	38.3234	12.9616	25.2612

It is obvious from Tables 1 and 2 that, robustness parameters NCC and NSR offering by a proposed algorithm 1 (using only PSO) is little more as compared to algorithm 2 (using PSO and AT both) against different types of noise attacks.

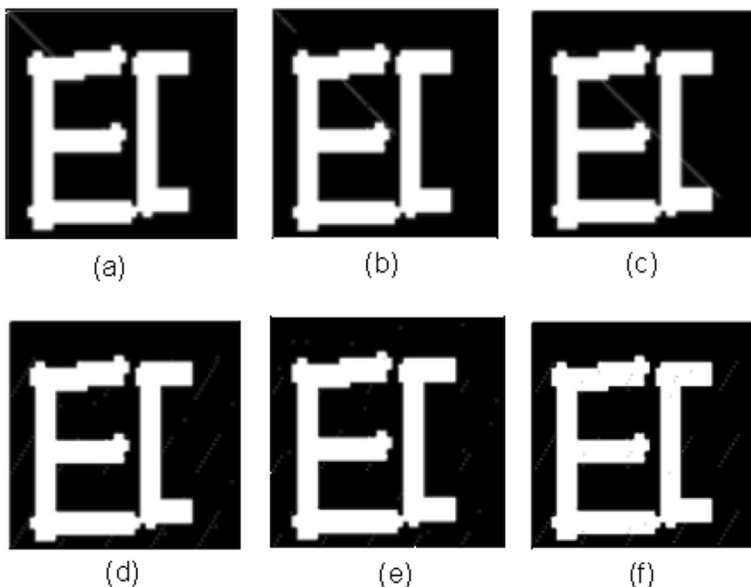
**Fig. 7** The extracted watermark images **a**, **b** and **c** using PSO based proposed algorithm **d**, **e** and **f** using AT-PSO based algorithm under attacks Histogram Equalization, Rotation and salt & pepper noise respectively

Table 3 Comparative analysis of NSR and PSNR values between existing and proposed techniques for Pepper cover image

Attacks	NSR		PSNR			
	Jane et al. [15]	Proposed		Jane et al. [15]	Proposed	
		PSO	PSO-AT		PSO	PSO-AT
Gauss. Blur	0.9946	1	0.9997	29.86	32.83	36.21
Scaling (50%)	0.9925	0.9983	0.9968	29.77	40.15	41.14
Hist. Equi.	0.988	0.9912	0.9928	17.54	17.33	17.32
Rot. (20°)	0.9883	0.9890	0.9950	11.40	11.49	10.73
JPEG (Q=25)	0.9929	0.9999	0.9979	8.33	32.95	36.65
Cropp. (12%)	0.9947	0.9971	0.999	13.04	19.06	13.72
Salt & Pepper	0.9076	0.9898	0.9954	12.33	25.25	25.11

However, the presence of diagonal line in Fig. 7a to c is showing limitation of PSO based algorithm 1. Also, use of AT along with PSO in proposed algorithm 2 is eliminate the problem of diagonal line in the extracted watermark image and offering NCC and NSR values closer to the algorithm 1. In addition, the PSNR value of noisy watermarked image in proposed algorithm 2 is better than algorithm 1 in case of most images in presence of variety of attacks.

Comparative analysis between proposed and existing scheme has been carried out and results are mentioned in Tables 3 and 4. The value of NSR using proposed algorithm is better than existing algorithm [15]. The performance of algorithm is more accurate and robust against Salt and Pepper image with their NSR values 0.9898 and 0.9954 respectively PSO and AT-PSO based algorithm. Moreover, PSNR value of noisy watermarked image using proposed and existing algorithm is depicted in Tables 1 and 3. Analysis of these characteristic show that PSNR of proposed algorithm is greater than existing algorithm in presence of any of the type of noise attack.

A comparison of proposed algorithm is also carried out with three different state-of-the-art algorithms and it is tabulated in Table 5. These algorithms are using optimization techniques like PSO and DE to embed watermark image into cover image. Here, a comparative analysis shows that against every reported attack both the proposed schemes are performed better as compared to all the existing watermarking algorithm. An analysis for PSNR Vs Payload is also

Table 4 Comparative analysis of NSR and PSNR values between existing and proposed techniques for Goldhill cover image

Attacks	NSR		PSNR			
	Jane et al. [15]	Proposed		Jane et al. [15]	Proposed	
		PSO	PSO-AT		PSO	PSO-AT
Gauss. Blur	0.9946	0.9999	0.9974	31.71	39.71	45.05
Scaling (50%)	0.9936	0.9985	0.9976	30.48	47.60	41.19
Hist. Equi.	0.9878	0.9972	0.9949	20.17	19.97	46.27
Rot. (20°)	0.9957	0.9961	0.9968	10.03	9.96	10.15
JPEG (Q=25)	0.9880	0.9997	0.9984	8.082	33.38	38.31
Cropp. (12%)	0.9947	0.9978	0.9961	12.87	15.88	14.59
Salt & Pepper	0.9047	0.9946	0.9953	12.05	22.19	25.09

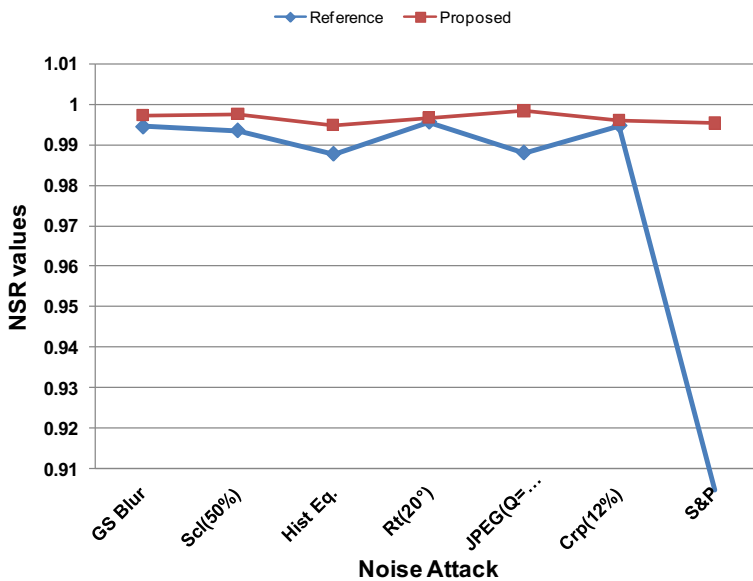


Fig. 8 The comparison of NSR values of extracted watermark of existing algorithm with the AT-PSO based proposed technique for Pepper cover image against different noise attacks

carried out with different size like 256×256 , 128×128 and 64×64 . With these analysis, Obtained PSNR values 30.5614, 32.3426 and 33.8793 sequentially for Goldhill Cover image with the size of 512×512 .

Many researchers have tried to improve the algorithm proposed in [15] using various optimization algorithm. Among them Bi-directional Extreme Learning Machine (B-ELM) based semi-blind watermarking is proposed in [24]. This algorithm is not only optimized RMSE but also scaling factor. The comparative analysis in Table 5 show that both PSO and AT-PSO algorithms both are performed better as compared to the algorithm given in [24].

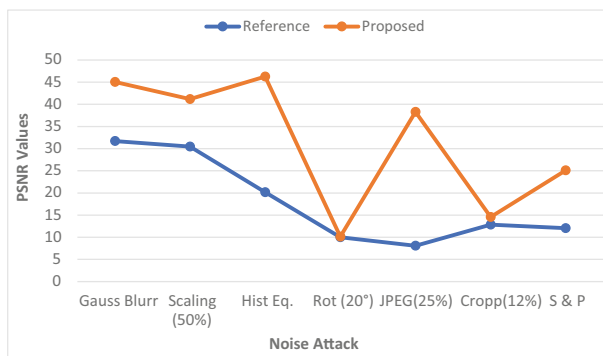


Fig. 9 The comparison of PSNR values of noisy watermarked image of existing algorithm with the AT-PSO based proposed technique for Pepper cover image against different noise attacks

Table 5 Comparative analysis of NCC values between existing and proposed techniques for Pepper cover image

Attacks	NCC				
	Yu et. Al. [35]	Ali et al. [1]	Mishra et al. [24]	Proposed	
				PSO	PSO-AT
Gauss. Blur	0.994	0.9906	0.96	0.9999	0.9980
Scaling (50%)	0.997	0.9731	0.98	0.999	0.9983
Hist. Equi.	NA	0.9654	NA	0.998	0.9964
Rot. (20°)	0.832	NA	NA	0.9973	0.9977
JPEG (Q=25)	0.883	0.9892	NA	0.9998	0.9988
Cropp. (12%)	NA	NA	NA	0.9984	0.9972
Salt & Pepper	0.851	NA	0.98	0.9962	0.9967

7 Conclusion

A reliable watermarking technique is proposed in [15] which is robust and imperceptible, but the procedure to select scale factor for embedding watermark is not adequate to get the maximum imperceptibility and robustness and limited up to a particular image. To make existing algorithm generalized for variety of images and to improve the imperceptibility and robustness, the PSO is incorporate. The comparative analysis of Tables 3 and 4 are showing that proposed PSO based approach not only increasing the robustness of algorithm but also the imperceptibility. In addition to that, Table 5 is also representing the better performance of the proposed algorithm as compared to existing algorithm [24] which is extended version of the algorithm used in [15].

The diagonal line problem of existing algorithm is not resolve by proposed approach based on PSO, therefore to remove this diagonal line from extracted watermark image the AT along with PSO is used. This AT-PSO based approach increases the robustness and imperceptibility as compare to existing algorithm. The reliability of existing algorithm is good and which further increases by use of AT in this paper.

Acknowledgements The authors would like to thank the anonymous referees for their valuable comments and suggestions that enhanced the content and the form of this paper.

References

1. Ali M, Ahn C (2014) An optimized watermarking technique based on self-adaptive DE in DWT-SVD transform domain. *Signal Process* 94:545–556
2. Aslantas V (2008) A singular-value decomposition-based image watermarking using genetic algorithm. *Electron Commun* 62(5):386–394
3. Bhatnagar G, Raman B (2009) A new robust reference watermarking scheme based on DWT-SVD. *Comput Stand Interfaces* 31(5):1002–1013
4. Chandra MB, Srinivas KS (2008) A robust image watermarking scheme using singular value decomposition. *J Multimed* 3(1):7–15
5. Cox I et al (1997) Secure spread spectrum watermarking for multimedia. *IEEE Trans Image Process* 6(12):1673–1687

6. Farah M, Farah A (2020) An image encryption scheme based on a new hybrid chaotic map and optimized substitution box. *Nonlinear Dyn*. <https://doi.org/10.1007/s11071-019-05413-8>
7. Ghazy R et al (2008) Performance evaluation of block based SVD image watermarking. *Prog Electromagn Res B* 8:147–159
8. Ghazy R et al (2015) Block-based SVD image watermarking in spatial and transform domain. *Int J Electron* 102(7):1091–1113
9. Ghosal S, Mandal J (2014) Binomial transform based fragile watermarking for image authentication. *J Inf Secur Appl* 19(4–5):272–281
10. Ghosal S, Mandal J (2019) On the use of the Stirling transform in image steganography. *J Inf Secur Appl* 46:320–330
11. Ghosal S et al (2020) Application of Lah transform for security and privacy of data through information hiding in telecommunication. *Trans Emerg Telecommun Technol*. <https://doi.org/10.1002/ett.3984>
12. Goyal S, Gupta R, Bansal A (2009) Application of genetic algorithm to optimize robustness and fidelity of watermarked images. *Comput Sci Eng* 1(3):239–242
13. Guo J, Prasetyo H (2014) False-positive-free SVD-based image watermarking. *Vis Commun Image Represent* 25(5):1149–1163
14. Jain C, Arora S, Panigrahi P (2008) A reliable SVD based watermarking scheme. Cornell University Library, arXiv:0808.0309v1 [cs.MM]. <http://arxiv.org/abs/0808.0309v1>
15. Jane O, Elbasi E, Ilk H (2014) Hybrid non-blind watermarking based on DWT and SVD. *J Appl Res Technol* 12(4):750–761
16. Jordehi A, Jasni J (2013) Parameter selection in particle swarm optimisation: a survey. *J Exp Theor Artif Intell* 25(4):527–542
17. Lai C-C (2011) An improved SVD-based watermarking scheme using human visual characteristics. *Opt Commun* 284(4):938–944
18. Lai C-C, Tsai C-C (2010) Digital image watermarking using discrete wavelet transform and singular value decomposition. *IEEE Trans Instrum Meas* 59(11):3060–3063
19. Li L, Gao N (2012) Research of a digital image watermarking algorithm based on DCT transform. Springer-Verlag, London
20. Liang R et al (2019) An improved Arnold image scrambling algorithm. *IOP Conf Series: Mater Sci Eng* 677:042020. <https://doi.org/10.1088/1757-899X/677/4/042020>
21. Lin C-C, Chang C-C, Chen Y-H (2014) A novel SVD-based watermarking scheme for protecting rightful ownership of digital images. *J Inf Hiding Multimed Signal Process* 5(2):124–143
22. Liu R, Tan T (2002) An SVD-based watermarking scheme for protecting rightful ownership. *IEEE Trans Multimed* 4(1):121–128
23. Makbol N, Khoo B (2013) A hybrid robust image watermarking scheme using integer wavelet transform, singular value decomposition and Arnold transform. In: Zaman HB, Robinson P, Olivier P, Shih TK, Velastin S (eds) *Advances in visual informatics. IVIC 2013. Lecture Notes in Computer Science*, vol 8237. Springer, Cham. https://doi.org/10.1007/978-3-319-02958-0_4
24. Mishra A, Rajpal A, Bala R (2018) Bi-directional extreme learning machine for semi-blind watermarking of compressed images. *J Inf Secur Appl* 38:71–84
25. Mohammad A, Alhaj A, Shaltaf S (2008) An improved SVD-based watermarking scheme for protecting rightful ownership. *Signal Process* 88(9):2158–2180
26. Pradhan C, Rath S, Bisoi A (2012) Non blind digital watermarking technique using DWT and cross chaos. *Procedia Tech* 6:897–904. <https://doi.org/10.1016/j.protcy.2012.10.109>
27. Rao R (2016) Jaya: a simple and new optimization algorithm for solving constrained and unconstrained optimization problems. *Int J Ind Eng Comput* 7:19–34
28. Rao S, Shekhavat R, Srivastava V (2012) A reliable digital image watermarking scheme based on SVD and particle swarm optimization. *Proc. Student conference on engineering and systems*, Allahabad, Uttar Pradesh, India. <https://doi.org/10.1109/SCES.2012.6199012>
29. Run R-S, Horng SJ, Lai JL, Kao TW, Chen RJ (2012) An improved SVD based watermarking technique for copyright protection. *Expert Syst Appl* 39(1):673–689
30. Rykaczewski R (2007) Comments on “an SVD-based watermarking scheme for protecting rightful ownership”. *IEEE Trans Multimed* 9(2):421–423
31. Shaamala A, Abdullah S, Manaf A (2011) Study of the effect DCT and DWT domains on the imperceptibility and robustness of genetic watermarking. *Int J Comput Sci Issues* 8(5):220–225
32. Shi H (2014) DWT and SVD based watermarking scheme with circulation. *J Softw* 9(3):655–662
33. Song J, Wang G (2011) A wavelet domain digital image-watermarking by double encryption based on Arnold transform and chaos. In: Chen R. (eds) *2011 International Conference in Electrics, Communication*

- and Automatic Control Proceedings. Springer, New York, NY. https://doi.org/10.1007/978-1-4419-8849-2_76
34. Wang Z, Sun X, Zang D (2007) A novel watermarking scheme based on PSO algorithm. Springer-Verlag, Berlin
 35. Yu C et al (2019) An adaptive and secure holographic image watermarking scheme. *Entropy* 21(5):460
 36. Zhang X, Li K (2005) Comments on “an SVD-based watermarking scheme for protecting rightful ownership”. *IEEE Trans Multimed* 7(2):593–594

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.